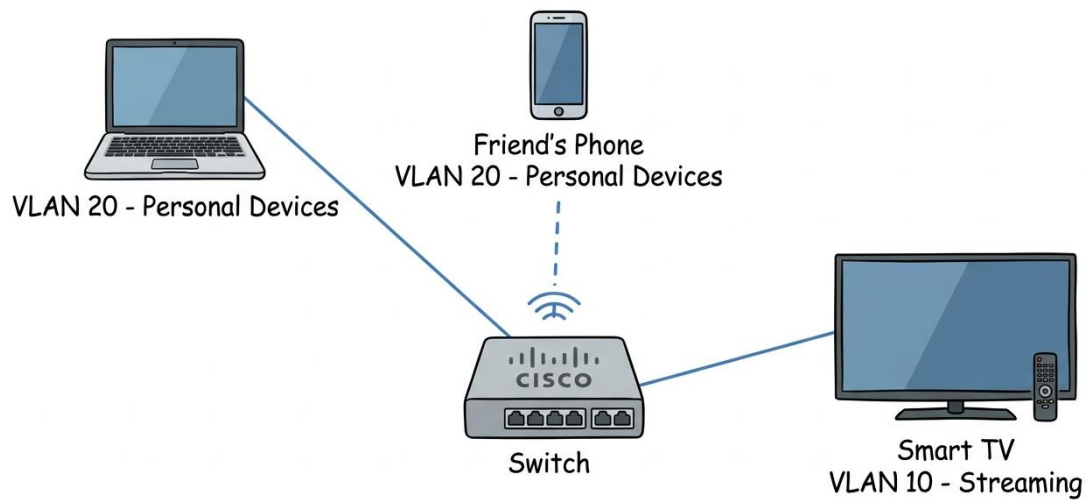


Task 1 — Network Diagram



VLAN 10 – Streaming: Smart TV
VLAN 20 – Personal Devices: Laptop,
Friend's Phone

VLANs help to separate devices into different logical networks. Devices in different VLANs normally cannot communicate directly through a Layer 2 switch.

Task 2 — Create VLAN 10 and VLAN 20

I used **Cisco Packet Tracer** for this task.

```
Switch> enable
```

```
Switch# configure terminal
```

```
Switch(config)# vlan 10
```

```
Switch(config-vlan)# name STREAMING
```

```
Switch(config-vlan)# exit
```

```
Switch(config)# vlan 20
```

```
Switch(config-vlan)# name PERSONAL
```

```
Switch(config-vlan)# exit
```

Assign 3 PCs to VLAN 10

For example, connect PC1, PC2 and PC3 to Fast Ethernet 0/1, 0/2 and 0/3.

```
Switch(config)# interface range fastEthernet 0/1-3
```

```
Switch(config-if-range)# switchport mode access
```

```
Switch(config-if-range)# switchport access vlan 10
```

```
Switch(config-if-range)# exit
```

```
Switch(config)# interface range fastEthernet 0/1-3
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)# exit
```

Assign 2 PCs to VLAN 20

Connect PC4 and PC5 to FastEthernet 0/4 and 0/5.

```
Switch(config)# interface range fastEthernet 0/4-5
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
Switch(config-if-range)# exit
```

Verify VLANs

```
Switch# show vlan brief
```

You should see:

```
VLAN 10  STREAMING  Fa0/1, Fa0/2, Fa0/3
```

```
VLAN 20  PERSONAL  Fa0/4, Fa0/5
```

Ping Test

PC1, PC2 and PC3 are in VLAN 10, so they can communicate with each other if their IP addresses are configured in the same subnet.

PC4 and PC5 are in VLAN 20, so they can communicate with each other.

PCs in VLAN 10 and VLAN 20 **cannot ping each other directly** because they are separated into different VLANs.

Result:

The VLAN configuration was successful and devices in different VLANs were isolated.

Task 3 — Move a PC from VLAN 10 to VLAN 20

Suppose **PC3** is connected to **FastEthernet 0/3**.

Before Configuration

PC3 is in VLAN 10:

```
Switch# show vlan brief
```

```
VLAN 10  STREAMING  Fa0/1, Fa0/2, Fa0/3
```

```
VLAN 20  PERSONAL  Fa0/4, Fa0/5
```

Move PC3 to VLAN 20

```
Switch# configure terminal
```

```
Switch(config)# interface fastEthernet 0/3
```

```
Switch(config-if)# switchport mode access
```

```
Switch(config-if)# switchport access vlan 20
```

```
Switch(config-if)# exit
```

After Configuration

```
Switch# show vlan brief
```

Now it should show:

```
VLAN 10  STREAMING  Fa0/1, Fa0/2
```

```
VLAN 20  PERSONAL  Fa0/3, Fa0/4, Fa0/5
```

Connectivity Test

Before moving PC3, it could communicate with PCs in VLAN 10.

After moving PC3 to VLAN 20, it can communicate with PCs in VLAN 20 but cannot directly communicate with PCs remaining in VLAN 10.

Result:

The PC was successfully moved from VLAN 10 to VLAN 20 and the connectivity changed according to the new VLAN.

Task 4 — Assign 10 Ports to VLAN 30

For this task I used **Cisco Packet Tracer**.

First create VLAN 30:

```
Switch> enable
```

```
Switch# configure terminal
```

```
Switch(config)# vlan 30
```

```
Switch(config-vlan)# name OFFICE
```

```
Switch(config-vlan)# exit
```

Now assign ports **Fa0/1 to Fa0/10** using an interface range command:

```
Switch(config)# interface range fastEthernet 0/1-10
```

```
Switch(config-if-range)# switchport mode access
```

```
Switch(config-if-range)# switchport access vlan 30
```

```
Switch(config-if-range)# exit
```

Verify

```
Switch# show vlan brief
```

The output should show:

```
VLAN 30  OFFICE  Fa0/1, Fa0/2, Fa0/3, Fa0/4,  
                Fa0/5,
```

Fa0/6, Fa0/7, Fa0/8, Fa0/9, Fa0/10

Result:

All 10 switch ports were assigned to VLAN 30 using the interface range command instead of repeating the configuration for every port.

Final Conclusion

VLANs are used to divide a network into separate logical networks. In this assignment, I created VLAN 10 and VLAN 20, assigned PCs to different VLANs, tested connectivity, moved a PC from one VLAN to another, and assigned 10 ports to VLAN 30 using an interface range command. This helped me understand how VLANs provide network separation and improve network organization.